

Problem statemen

Create a program, using the bubble sort algorithm, to sort an array of integers.

Download and copy the files **bubble_sort.txt** and **num.txt** in your drive.

Code:

- Must use only the **bubble sort**. (use file provided: **bubble_sort.txt**)
- Modify it to stop as soon as the array is sorted.
- The array is sorted if there are no swaps on a pass through an array.
- Prompt user for the fully qualified path name of the file.(*Enter full path for the input file and the output file*)
- Use the sample file **num.txt** containing **500 integers**.
Your program will read from this input file.
- The sorted array must be printed to an **output file**. User choses filename.
The output file should display a single integer on each line.
The last line of the output file **must contain the count of number of times integer values were swapped**.
- Open the file in a function – then **error check** for successful open.
- **Use following functions:** - **get_input_fileName**, **get_output_file_Name**,
bubble_sort_method, **display_sorted array**.

You are free to make additional functions for your code.

Use the comment header and function headers for your source code.

You are to create a project/file for this lab named **Lab8** for the project name and **lab8_driver.cpp** for the driver.

Submission Guidelines:

Browse to your Lab Submissions module in course tab **CS 121**

- Upload the source code **Lab8.cpp** and the **text file** containing the screen capture of the output window.
- You must submit a source code Lab8.cpp, text document containing **screen captures** of your running program.